

Frequency Response Modeling

EE-5360 Renewable Energy Sources

Angelo Torres

Dr. Jamehbozorg

10/3/2026

Introduction

An increase in demand draws on the kinetic energy stored in rotating generators, decreasing system frequency. Governor and frequency-sensitive load responses help restore the power balance. This project graphs the released demand, additional governor power, and frequency for 50 seconds following the increase in demand from 70 to 73 GW. Then it examines the effects of increased droop and reduced inertia.

Method

The system has a maximum capacity of P_{max} = 90 GW, with the starting generation level set at $P_{G,0}$ = 70 GW, while the demand following a step change is at P_L = 73 GW at a standard frequency of f_{sp} 60 Hz. The nominal frequency f_{nom} and the governor setpoint f_{sp} are both maintained at 60 Hz. The load-frequency sensitivity is D = 0.3, and the governor reaction coefficient stands at G = 0.3.

In part (a), we are considering a droop of 0.04, along with an inertia constant of H = 6 sec. For part (b), we will simply modify the droop to 0.06. Then in part (c), we return the droop to 0.04, but update the inertia to 4.8 seconds. We need to evaluate several state variables: frequency, kinetic energy, and the additional governor power exceeding the initial 70 GW. P_{tar} denotes the governor target, P_R indicates the demand that has been released, and P_S is the additional generation. I represents the equivalent rotational inertia. The time step is denoted by k, and we are conducting calculations in 0.001-second intervals, using watts, joules, seconds, and hertz as our units.

1. Initialize each case:

At t = 0, set f = 60 Hz and P_{Gov} = 0. Released demand is initially zero, so P_S = 70 + 0 + 0 − 73 = −3 GW. Compute the initial kinetic energy and equivalent inertia:

$$KE_0 = HP_{max},\ I = \frac{2HP_{max}}{(2\pi f_{nom})^2}$$

For H = 6 s, KE_0 = 6 × 90 = 540 GJ and I ≈ $7.599 \times 10^6\ kg{\cdot}m^2$. For H = 4.8 s, KE_0 = 4.8 × 90 = 432 GJ and I ≈ $6.079 \times 10^6\ kg{\cdot}m^2$.

2. Calculate power at the current time step:

Use the current frequency to calculate the governor target and release demand. Released demand is considered positive when it falls below 60 Hz and negative when it exceeds 60 Hz; the actual demand is simply P_L minus P_R. Following this, you can compute the surplus power:

$$P_{tar,k} = \frac{f_{sp} - f_k}{R f_{nom}} P_{max}$$

$$P_{R,k} = -DP_L \frac{f_k - f_{nom}}{f_{nom}}$$

$$P_{S,k} = P_{G,0} + P_{Gov,k} + P_{R,k} - P_L$$

3. Update governor power, energy and frequency:

Advance the governor's power and kinetic energy by using the forward-Euler technique. Both updates use the current time step k values. Once the energy is updated, determine the next frequency based on the new energy.

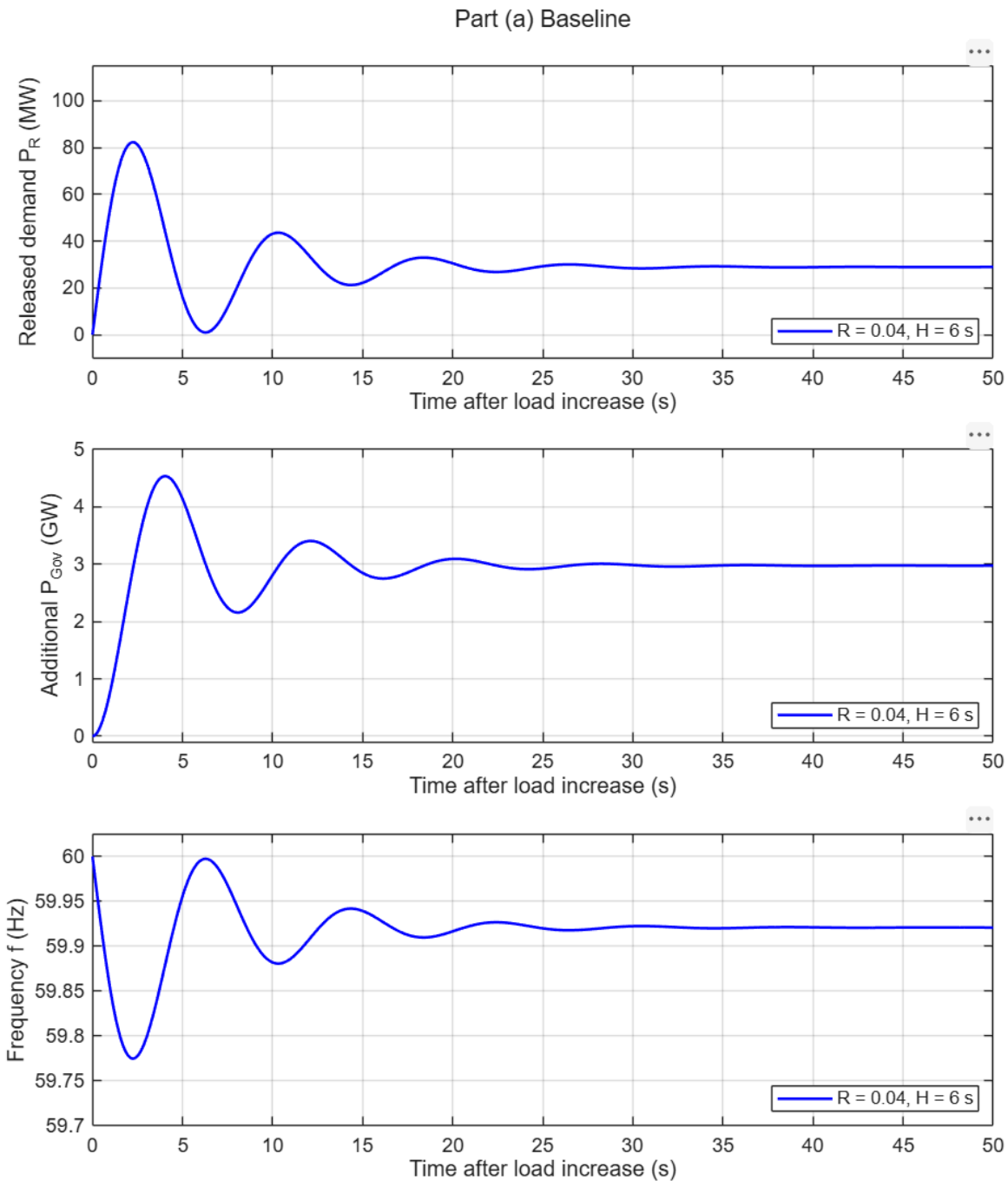
$$P_{Gov,k+1} = P_{Gov,k} + (P_{tar,k} - P_{Gov.k})G\Delta t$$

$$KE_{k+1} = KE_k + P_{S,k}\Delta t,\ f_{k+1} = \frac{1}{2\pi}\sqrt{\frac{2KE_{k+1}}{I}}$$

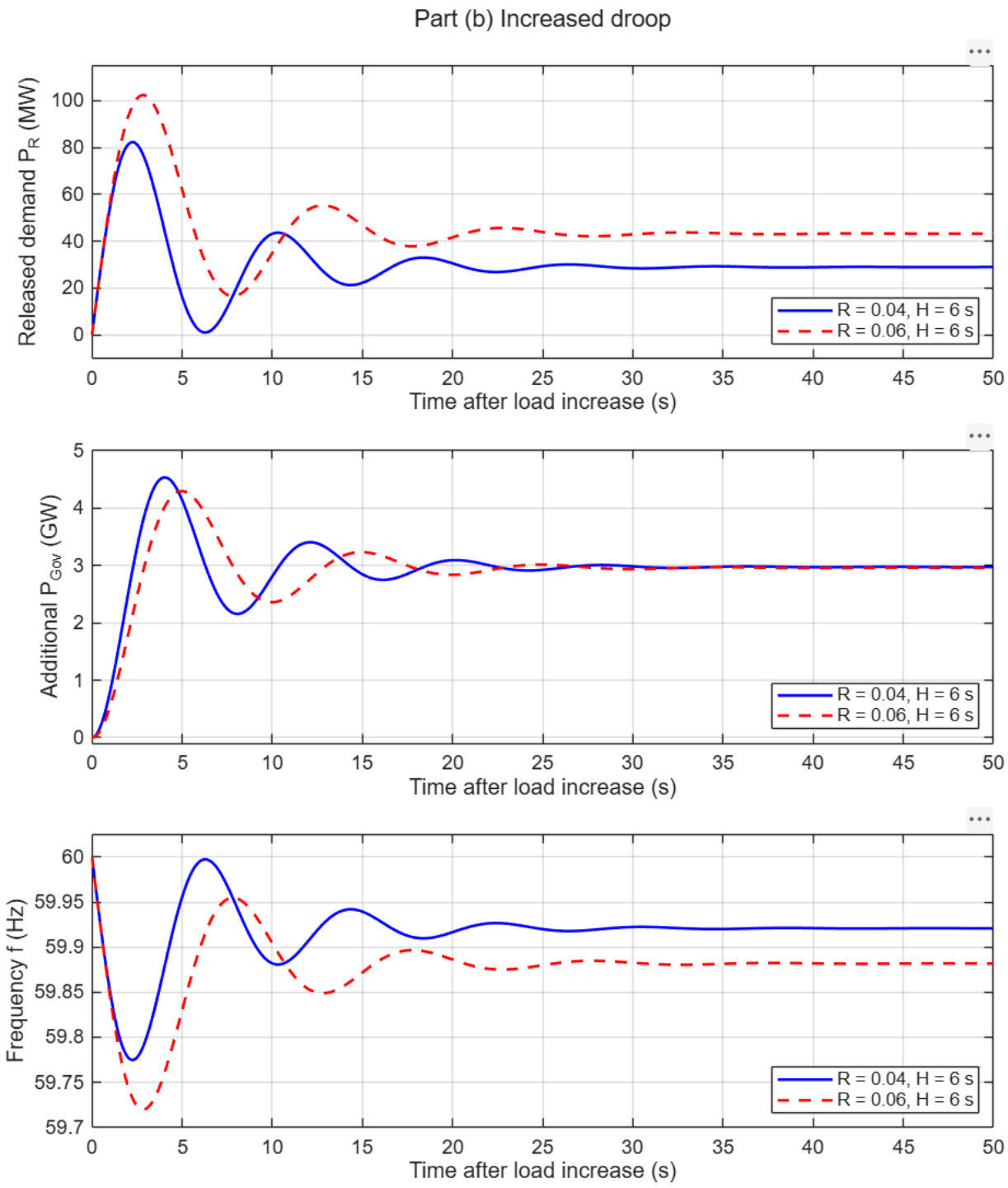
For the first baseline step, we set P_{tar} to 0 and keep P_{Gov} at zero too. This means the energy is calculated as $540 \times 10^9 - 3 \times 10^9(0.001) = 539.997 \times 10^9\ J$. From this, we get a frequency of roughly 59.999833 Hz. This slight frequency error triggers the governor response in the next steps.

Repeat 2 and 3 until t = 50 s for each case. Plot P_R in MW, P_{Gov} in GW, and f in Hz. For parts b) and c), overlay each changed case with the baseline on the same axes.

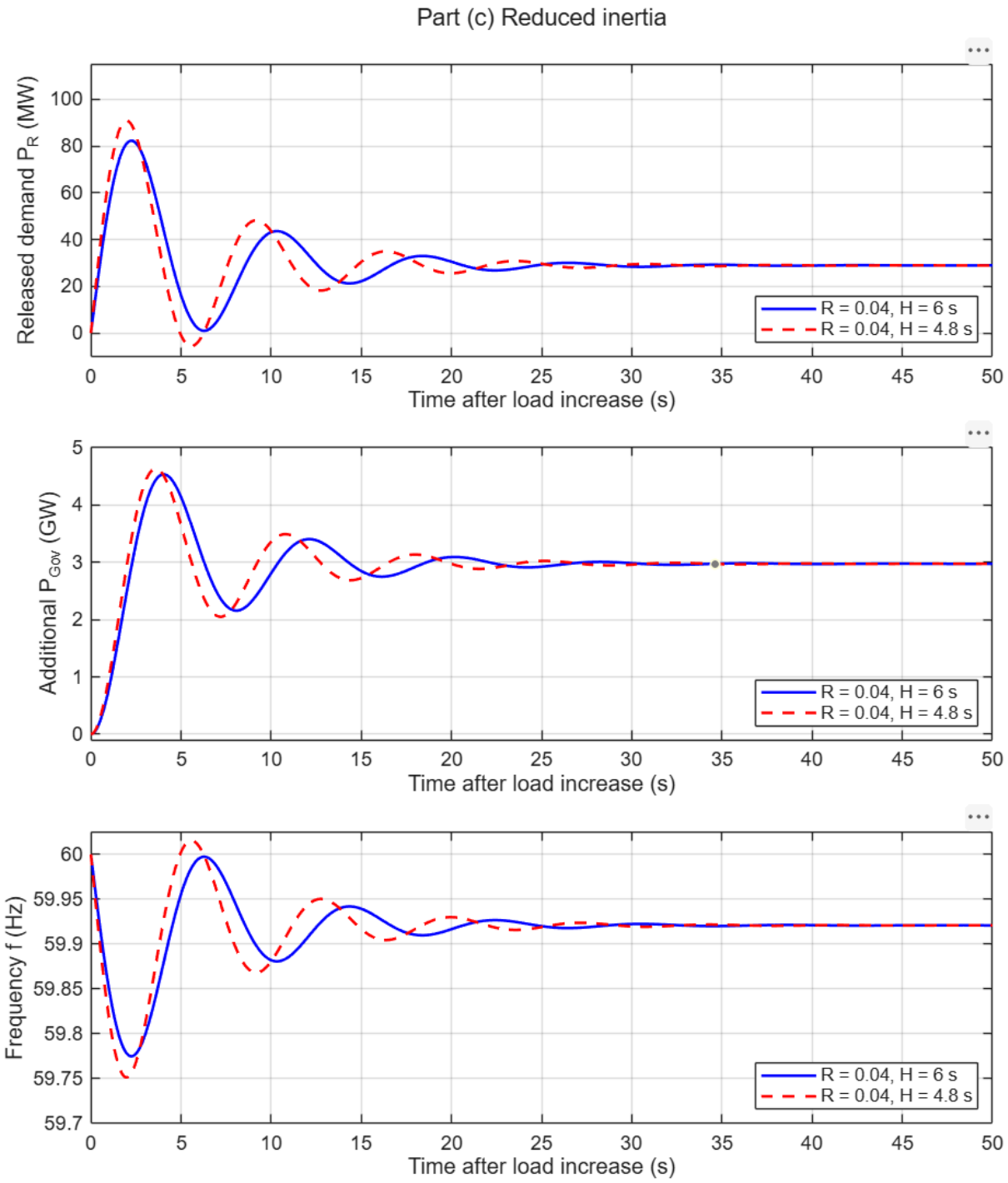
Results



Baseline: R = 0.04 and H = 6 s. The three panels show released demand, additional governor power, and frequency over 50 seconds.



Case A (blue solid): R = 0.04. Case B (red dashed): R = 0.06. Both use H = 6 s.



Case A (blue solid): H = 6 s. Case C (red dashed): H = 4.8 s. Both use R = 0.04.

Discussion/Conclusion

For part a, the first 3 GW shortfall cuts down stored energy and lowers the frequency. At around 2.245 seconds, the frequency drops to about 59.775 Hz and then moves closer to 59.921 Hz. When the frequency dips, demand goes up right away, but the governor power reacts after a

bit. The highest governor power hits around 4.530 GW before the fluctuations die down. At equilibrium, about 2.971 GW of extra generation plus 28.92 MW of released demand make up for the 3 GW shortfall, just like Equation (4) shows.

For part b, A bigger droop means the governor target drops for the same frequency error according to Equation (2). With R at 0.06, the frequency drops to roughly 59.720 Hz at 2.836 seconds and gets closer to 59.882 Hz. The peak governor power is lower and comes in later, peaking at about 4.290 GW at 4.978 seconds, in contrast to 4.530 GW at 4.033 seconds in the baseline. The bigger drop in frequency raises the peak released demand from 82.22 MW to 102.30 MW. When it all settles, governor power is a bit lower at 2.957 GW, while released demand goes up to 43.17 MW; their total still balances out the 3 GW increase.

For part c, less inertia reduces the initial stored energy from 540 GJ to 432 GJ. Because of this, the frequency falls faster and hits a deeper low of around 59.751 Hz at 1.976 seconds. The bigger early frequency error pushes the peak released demand up to 90.85 MW and peak governor power to 4.629 GW. A short overshoot in frequency above 60 Hz makes the released demand go negative, which shows a tiny rise in consumption. All three figures get closer to the baseline because R and D don't change; inertia only affects how energy reacts in the short term, not the final power balance. More droop makes the steady-state frequency error worse, while less inertia messes with the initial frequency response. The frequency stays below 60 Hz since primary droop control needs a frequency error to keep governor power up, and there's no secondary frequency restoration in the model.

Appendix

```
clear; clc; close all;

Pmax = 90e9; Pg0 = 70e9; PL = 73e9; % W

D = 0.3; fnom = 60; fsp = 60; G = 0.3;

R = [0.04 0.06 0.04]; % Cases a, b, c

H = [6 6 4.8];

dt = 0.001;

t = (0:dt:50)'; N = length(t);

f = zeros(N,3); Pgov = zeros(N,3);

f(1,:) = fnom;

% Pgov is additional generation above the original 70 GW.

for j = 1:3
```

```
  KE = H(j)*Pmax;
  I = 2*KE/(2*pi*fnom)^2;
  for k = 1:N-1
     Ptar = (fsp-f(k,j))*Pmax/(R(j)*fnom);
     PRk = -D*PL*(f(k,j)-fnom)/fnom;
     Ps = Pg0 + Pgov(k,j) + PRk - PL;
     % Both updates use the current states at index k.
     Pgov(k+1,j) = Pgov(k,j) ...
        + (Ptar-Pgov(k,j))*G*dt;
     KE = KE + Ps*dt;
     f(k+1,j) = sqrt(2*KE/I)/(2*pi);
  end
end
PR = -D*PL*(f-fnom)/fnom;
% Part a: baseline. Part b: a versus b. Part c: a versus c.
groups = {1, [1 2], [1 3]};
titles = {'Part (a) Baseline', 'Part (b) Increased droop', ...
       'Part (c) Reduced inertia'};
labels = {'R = 0.04, H = 6 s', 'R = 0.06, H = 6 s', ...
       'R = 0.04, H = 4.8 s'};
values = {PR/1e6, Pgov/1e9, f};
ylabels = {'Released demand P_R (MW)', ...
        'Additional P_{Gov} (GW)', 'Frequency f (Hz)'};
limits = {[-10 115], [-0.1 5], [59.70 60.025]};
styles = {'b-', 'r--'};
outdir = fullfile(fileparts(mfilename('fullpath')), 'plots');
if ~exist(outdir,'dir'), mkdir(outdir); end
```

```
for g = 1:3
  figure('Color','w','Position',[100 100 850 950]);
  tiledlayout(3,1,'TileSpacing','compact');
  for panel = 1:3
    nexttile; hold on;
    for q = 1:length(groups{g})
      j = groups{g}(q);
      plot(t,values{panel}(:,j),styles{q}, ...
        'LineWidth',1.5,'DisplayName',labels{j});
    end
    xlabel('Time after load increase (s)');
    ylabel(ylabels{panel});
    xlim([0 50]); ylim(limits{panel});
    grid on; box on;
    set(gca,'FontSize',11);
    legend('Location','southeast');
  end
  sgtitle(titles{g});
  filename = sprintf('part_%c.png','a'+g-1);
  exportgraphics(gcf,fullfile(outdir,filename),'Resolution',300);
end
```